INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

TESTER, DENSITY AND MOISTURE (SOIL AND ASPHALT), NUCLEAR METHOD

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1 SCOPE. This commercial item description covers a single unit (radioactive) source density and moisture tester. The tester is used for measurement of density and moisture content of soils, base materials, and density of asphaltic concrete.

2 SALIENT CHARACTERISTICS. The tester shall be a single unit measuring device comprised of radioactive sources, radioactive detectors, a readout device, data storage, integral power supply, housing, and locking mechanism. The tester shall be capable of testing for density and moisture by backscatter, direct transmission, and (optional) air gap methods. Depth of measurement shall be up to 8.00 inches in 2-inch increments.

2.1 Radioactive sources. The sources shall be encapsulated and sealed. The radiological surface dose rate shall not be greater than 15.0 millirem/hour (mrem/hr) or the radiation level at 12 inches from any surface with the source in the shielded position shall not be greater than 10.0 mrem/hr (neutron and gamma sources). Radium shall not be used as a radioactive source.

2.2 Radioactive detectors. The detectors shall be any suitable type of radioactive detectors.

2.3 Readout device. At a minimum, the readout device shall contain counting circuits, timing circuits, a data display, and related electronic components. The display shall be a direct readout of wet density, dry density, and moisture content, in pounds per cubic foot (pcf) with automatic conversion to metric units. The display shall clearly indicate the quantity and units of measure, in any ambient light.

Comments, suggestions, or questions on this document should be addressed to: U.S. ARMY TOPOGRAPHIC ENGINEERING CENTER, ATTN: CEERD-TS-T, 7701 TELEGRAPH ROAD, ALEXANDRIA, VA 22315-3864 or emailed to pd-te@tec.army.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

AMSC N/A FSC 6635

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2.4 Data storage. The tester shall have an internal data storage capability of not less than 120 test records, and the ability to transfer this to an electronic printer or computer. Each record shall include the date conducted and provision for entering a task identifier of not less than five characters.

2.5 Power supply. The power supply shall provide sufficient power to conduct not less than eight hours of normal, consecutive, 60-second measurements, prior to a recharge or replacement. If batteries are used, they shall be of a type that minimizes the promotion of corrosion, when left installed for extended periods.

2.6 Voltage indicator. An indicator or other device shall provide a clear distinction between when the voltage is adequate for testing and when on low reserve. In low reserve, the system shall be capable of not less than two hours of consecutive uses, prior to recharge or replacement.

2.7 Housing. The power supply, sources, detectors, readout device, etc., shall be in a housing of rugged construction that is moisture and dust proof. The housing shall conform to the U.S. Nuclear Regulatory Commission requirements in Title 10 of the Code of Federal Regulations, consumer product requirements and applicable ASTMs. When required, radiation labels mandated for the transportation case (see 2.8.5) shall also be applied to the housing.

2.8 Tool and accessories. The tester shall be furnished with a reference standard, all tools necessary for operation, site-preparation devices, technical publications, signs and postings**,** a battery charger, a locking mechanism and a transport case with locking mechanism.

2.8.1 Reference standard. The reference standard shall be a combination type, provided to check equipment operation, background count, and to establish conditions for a reproducible density and moisture count rate.

2.8.2 Site preparation devices. The devices shall provide for the following tests to comply with performance requirements specified in 2.9.

2.8.2.1 Backscatter. Site-preparation devices for the backscatter test method shall include a flat plate straightedge or other suitable tools, to be used to level the test site to the required uniformity.

2.8.2.2 Direct transmission. When required, the provided site-preparation devices for the direct transmission test method shall include an auger, rod, or other hole-forming device with marks to indicate depth. It shall have a bore diameter equal to or up to .12 inch larger than the probe, with a guide to ensure that the hole is perpendicular to the test site surface.

2.8.2.3 Air gap (optional). The provided site-preparation devices for the (optional) air gap test method shall include a cradle or spacers, to support the gage at the optimum gap above the material being tested.

2.8.3 Battery charger. An automatic charger with quick-disconnect plug-in type connectors shall be provided. The time it takes to fully recharge the battery shall not be greater than sixteen hours. The charger shall operate from 12 volts, dc or 110 volts, ac, 60-hertz power, and shall protect the voltage regulator from damage when the battery is connected improperly. The tester shall incorporate a battery overcharge prevention feature.

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2.8.4 Technical publications. Technical publications shall include supplier's standard operating, maintenance, safety, and parts manuals.

2.8.5 Transport case. The transport case shall contain the tester, all tools, accessories, technical publications, signs and postings, and a locking mechanism. The case shall conform to the U.S. Department of Transportation requirements in Title 49 of the Code of Federal Regulations. The exterior of the case shall contain all labels/markings as required by the regulations.

2.9 Performance. The tester shall meet the performance requirements of ASTM Standards D2922, D2950, and D3017. The tester shall also meet the following requirements.

2.9.1 Direct transmission.

Range of calibration	80 to 160 pounds per cubic foot (pcf)
Precision (6-inch depth) (at 120 pcf)	± 0.25 pcf
Chemical error (at 120 pcf)	± 1.50 pcf
Surface roughness error (0.05 in.)	– 0.90 pcf

2.9.2 Backscatter.

Range of calibration	80 to 160 pcf
Precision (6-inch depth) (at 120 pcf)	± 0.50 pcf
Chemical error (at 120 pcf)	± 2.50 pcf
Surface roughness error (0.05 in.)	– 4.00 pcf

2.9.3 Moisture.

Range of calibration	0 to 40 pcf
Precision (at 10 pcf)	± 0.25 pcf

2.9.4 Operational alarm. The tester shall emit an auditory pitch or sound when it is in its operational mode (when the radioactive source is in the unshielded position or operational).

2.10 Environmental conditions. The tester shall operate as specified herein in any ambient temperature from 32 ºF to 140 ºF. The tester shall operate as specified herein on hot materials (asphalt) at temperatures up to 350 F without the use of additional insulating material between the instrument and the surface being tested.

2.11 Weight. The tester weight shall not be greater than 42 pounds. The weight of the tester and case (with tools, publications, and accessories) shall not be greater than 116 pounds.

2.12 Serviceability. Serviceable subassemblies shall be removable without exposing the sources or disturbing their shielding assemblies.

2.13 Marking. The tester housing and transport case shall be marked with the unit serial and model numbers, National Stock Number (NSN), the type of radioactive sources and date of manufacture, the amount of radioactive material, and radiation caution symbol(s).

2.14 Workmanship. All parts shall be free from rust, tool marks, pits and other injurious defects. External surfaces shall be free of burrs, sharp edges and corners except where sharp edges or corners are required or where they will not be detrimental to safety. Electronic components shall be protected against adverse effects of moisture and fungus growth. Each unit

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shall undergo burn-in of sufficient duration to ensure sustained precision over the life of the tester.

3 REGULATORY REQUIREMENTS

3.1 Licensing and control of radiation sources. The contractor shall provide evidence that the tester has met the product safety requirements of the U.S. Nuclear Regulatory Commission for products containing radioactive sources, consumer product requirements, and provide the necessary data for licensing. The contractor shall furnish the contracting officer a copy of the license(s) required prior to shipment date, to comply with the requirements of 2.7 and 2.8.5.

3.2 Recovered material. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4 QUALITY ASSURANCE PROVISIONS

4.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance as provided for in the provisions of the contract or purchase order.

4.2 Market acceptability. The tester offered shall be the latest model of the standard product of the supplier and shall have been used in the field for at least one year by professional organizations such as State Highway Departments.

5 PACKAGING. Packaging. Preservation, packing, and marking shall be as specified in the contract or order.

6 NOTES. Intended use. The tester is intended measurements of moisture content and level of bulk and dry density compaction of soil, aggregate, concrete, and asphalt materials. The system provides data, in real time, to determine the suitability of sites for emplacement of materiel or construction, and sufficiency of trenches, tarmac, pavement, etc to support troop deployment or vehicular traffic.

6.2 Data requirements. The procuring agency should specify the level of preservation, packing, and marking required. The procuring agency may also need to specify inclusion of a gamma detection meter (monitor), when it is a requirement by the user's local regulatory agencies and /or federal agencies. The procuring agency should also specify that the manufacturer provides NRC Sealed Source (Special Form) for tester radioactive sources and DOT 7A certification for transport case, contractor certification of compliance, radioactive material caution signs, Energy Reorganization Act of 1974 – Section 206, the latest version of 10 CPR Parts 19, 20 and 21, and the latest version of NRC Form 3 with each tester. The procuring agency should specify that the transport case be moisture and dust resistant and have corrosion resistant hardware.

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6.3 Sources of Documents.

6.3.1 ASTM. ASTM D2922, D2950 and D3017 may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103 (E-mail: http//www.ASTM.org).

6.3.2 Regulations. Federal Acquisition Regulations (FAR) and Code of Federal Regulations (CFR) may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402 (E-mail: http://www.access.gpo.gov/).

6.4 National Stock Numbers (NSNs). The following NSN is assigned to a product that conforms to this CID. This may not be indicative of all possible NSNs associated with the CID.

NSN
6635-01-030-6896

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